

Core: Business Case

Chapter 3:
Requirements in a Project Framework

Chapter Objectives

In this chapter, we will position requirements in the project framework by:

- Establishing the work involved in project management
- Establishing a requirements life cycle
- Identifying software development life cycles
- Distinguishing between key project roles and responsibilities
- Managing scope and scope change

Requirements in a Project Framework



Project Management

Roles and Responsibilities

Life Cycles

Managing Scope

The Business Case Is Approved!

What's Next?

- An approved business case may become a project
 - Project manager (and optionally other resources) are assigned
 - Planning begins
 - When planning is done, work begins
 - In project management, “First you plan your work, then you work your plan”



Projects Defined



- The goal of a project is to create or enhance a product or a service
 - Intent is to provide value for the customer
- So what's a project?
 - Temporary endeavor
 - Definite starting point
 - Definite ending point
 - Unique goal-driven deliverable(s)
 - Normally taken from the Business Case

What Projects Are Not

- Projects are not:
 - Repetitive work efforts like “Updating the content on a web page”
 - Permanent activities like “Human Resources”
 - Semi-permanent assignments like “Seasonal Help”

PROJECT CENSORED

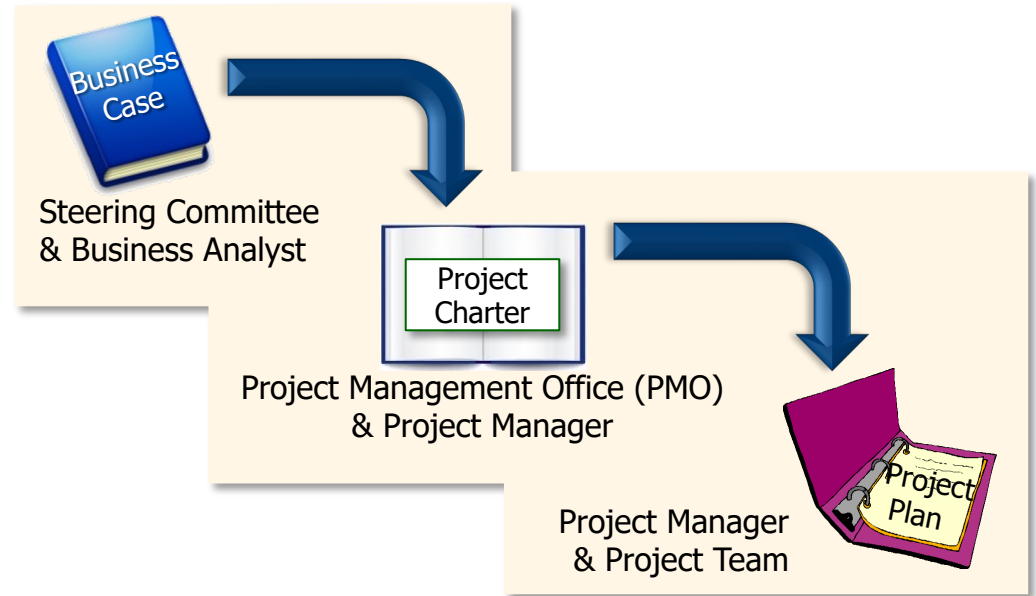
The News That Didn't Make The News

Project Management Plan

■ Project Manager is responsible for creating and following a plan

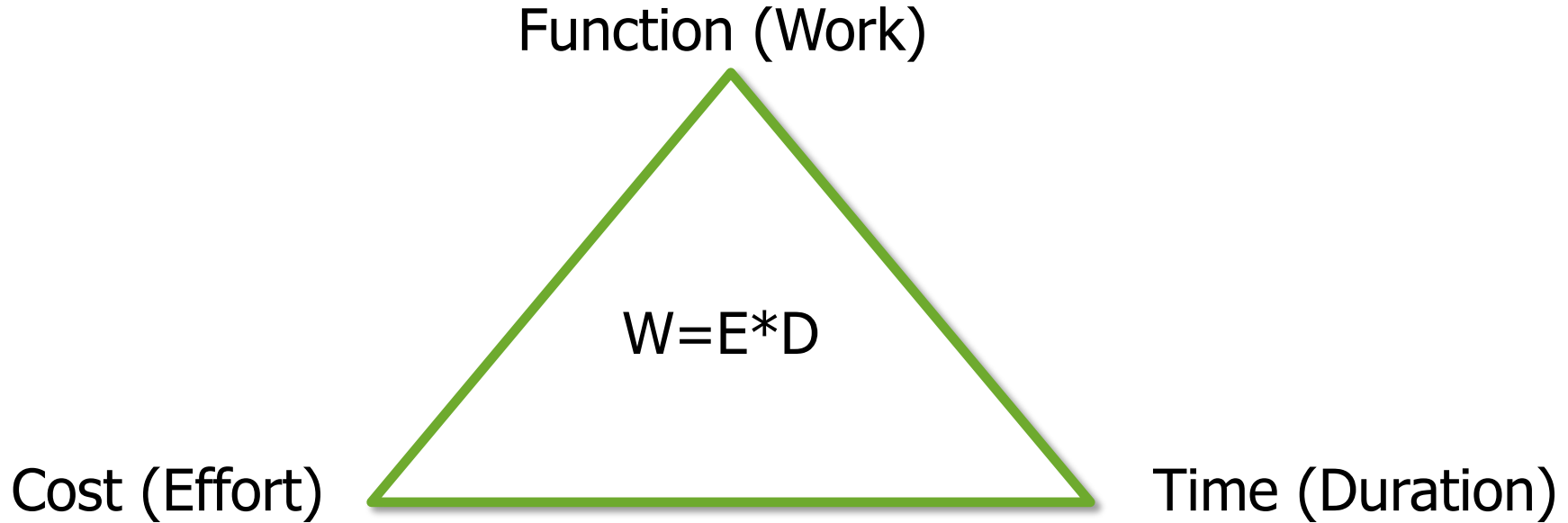
■ The plan is created during project initiation and covers:

- Scope
- Time Estimate
- Cost
- Quality - Quality Plan Artifact
- Human Resource Estimates
- Communications
- Risk
- Integration
- Procurement
- Stakeholder Management



It's All About the Triangle

- Project Management is about balancing Function, Cost, and Time
 - They are related, change one and change the others



Chaos in Projects

- Standish Group publishes the CHAOS study reporting on I.T. projects tracking success, challenges, and failures evaluating on characteristics including:
 - Project budgets
 - Project costs
 - Delivered functionality vs. expected functionality
- Success rates have barely moved in more than 25 years



Standish "Chaos Study" by Year							
Project Outcome	1994	2000	2002	2004	2009	2012	2020
Succeeded	16%	28%	34%	29%	32%	37%	31%
Challenged	53%	49%	51%	53%	44%	42%	50%
Failed	31%	23%	15%	18%	24%	21%	19%

Instructor-Led Activity: Sort the Top Factors Contributing to Challenged Projects



Factor	Percentage
Changing Requirements & Specifications	
Incomplete Requirements & Specifications	
Lack of Executive Support	
Lack of Resources	
Lack of User Input	
New Technology	
Technology Incompetence	
Unclear Objectives	
Unrealistic Expectations	
Unrealistic Time Frames	

12.8% 12.3% 11.8% 7.5% 7.0% 6.4% 5.9% 5.3% 4.3% 3.7%

Source: The Standish Group "CHAOS Summary 2009"

Instructor-Led Activity: Sort the Top Factors Contributing to Challenged Projects (continued)



Factor	Percentage
Changing Requirements & Specifications	11.8%
Incomplete Requirements & Specifications	12.3%
Lack of Executive Support	7.5%
Lack of Resources	6.4%
Lack of User Input	12.8%
New Technology	3.7%
Technology Incompetence	7.0%
Unclear Objectives	5.3%
Unrealistic Expectations	5.9%
Unrealistic Time Frames	4.3%

12.8% 12.3% 11.8% 7.5% 7.0% 6.4% 5.9% 5.3% 4.3% 3.7%

Source: The Standish Group "CHAOS Summary 2009"

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Who Is Involved?

- Two sets of individuals to consider
 - Those with a vested interest in the project
 - Stakeholders
 - Those with a vested interest in the SDLC
 - Team
 - Each group has its own focus and interests
 - Stakeholders: How will this project impact me?
 - Team: How will this project get done?
 - Above is true regardless of the SDLC selected



Stakeholders vs. Development Team



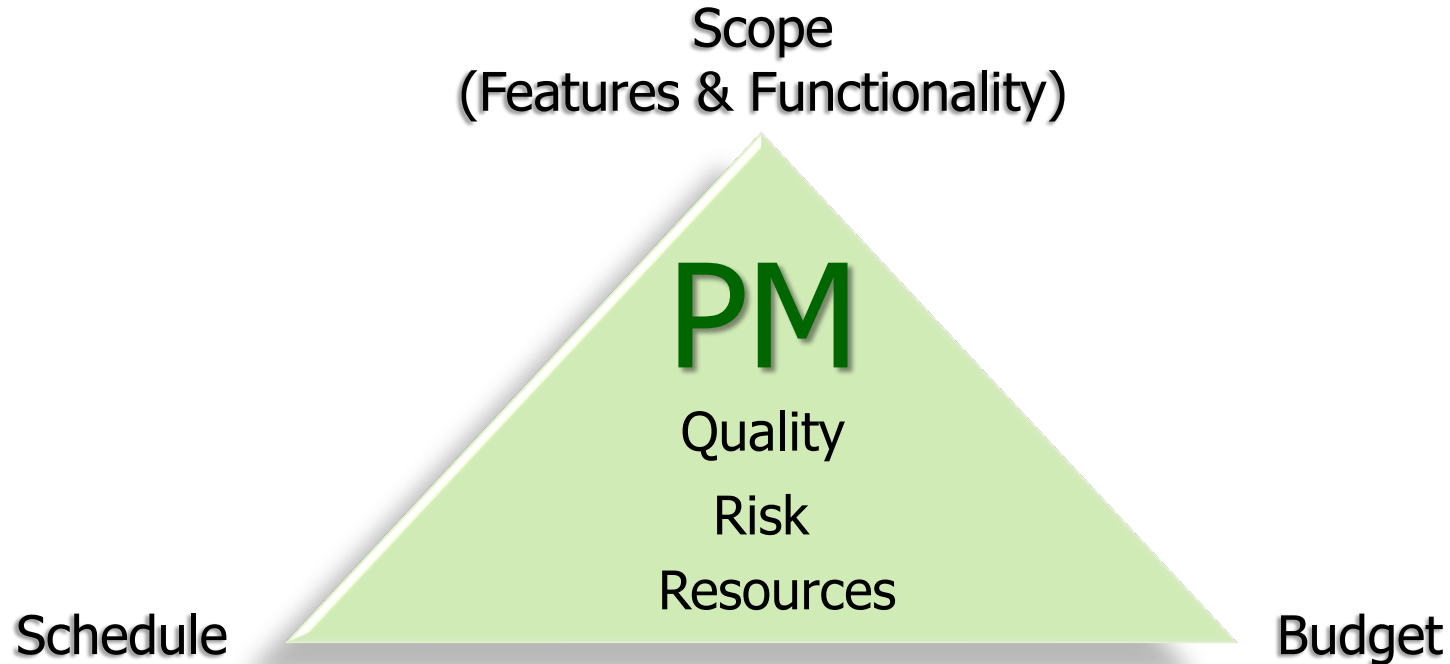
- Stakeholders
 - Project sponsor, security, finance, human resources, etc.
- Team Participants
 - Analysts, designers, programmers, etc.
- Project Manager belongs to both groups
 - What about team members?

Activity 3.1: Team and Stakeholder Roles

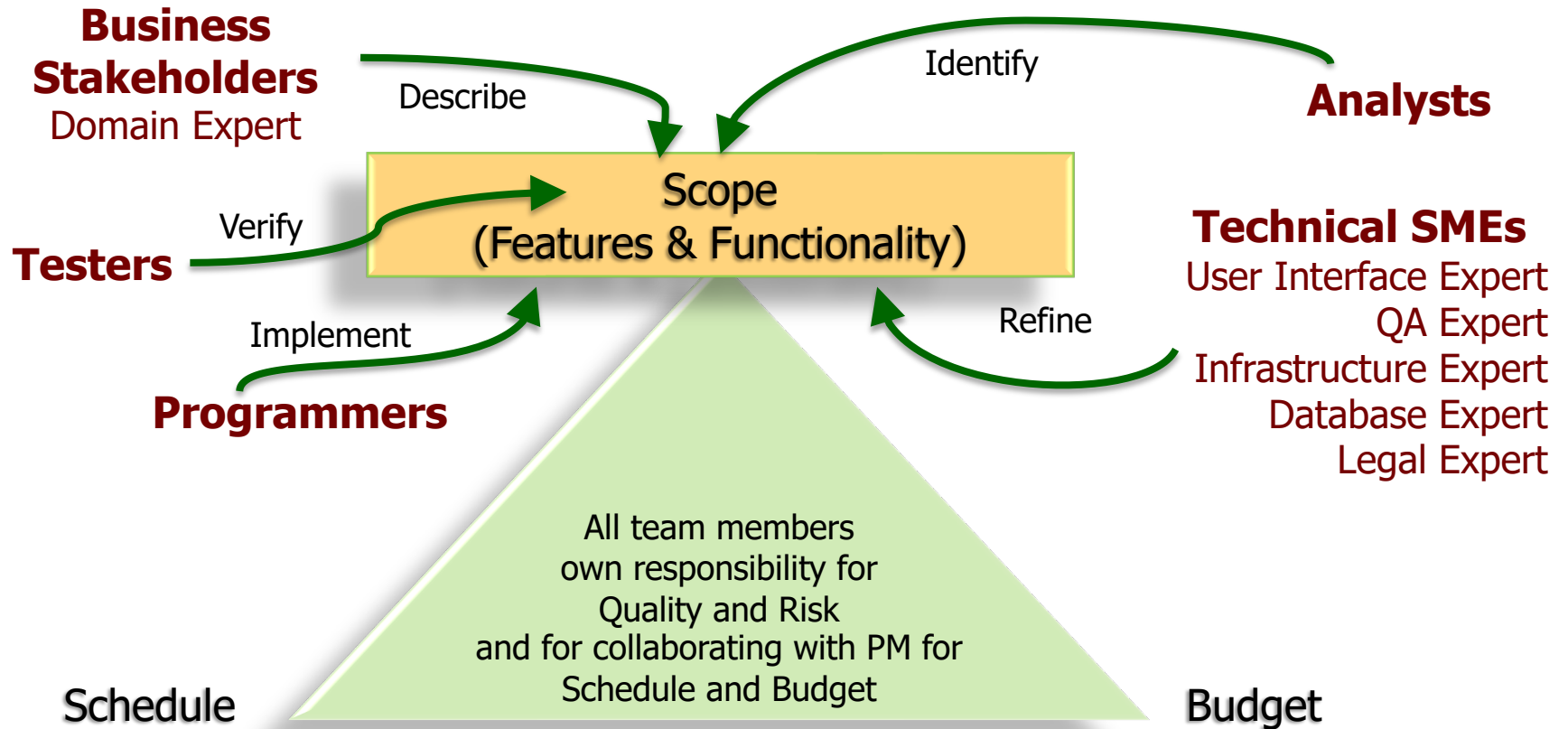


Goal	Identify the roles necessary in a typical software development project
Team Format	Case study teams
Materials	Flip chart
Tasks	1. Identify the roles of team members that you believe will be needed to produce the Buy Local Bonds application as a product of Meridian Financial 2. Identify as many stakeholders as possible with a vested interest in BLB 3. Prepare a flip chart with your answers
Deliverable	List of project roles
Duration	20 minutes

The Project Manager Role



Project Team Roles



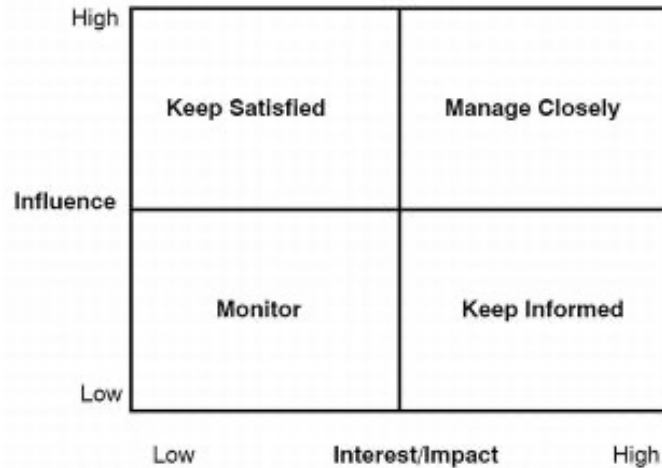
SME = Subject Matter Expert

Stakeholders

- Stakeholders are defined as individuals and organizations who are:
 - Actively involved in the project
 - Whose interests may be positively or negatively affected by either
 - The resulting product of the project
 - The execution of the project or project completion



Stakeholders (continued)



- Relative to the project, a stakeholder can be described as being:
 - Involved and working on the project in some way
 - Affected by the project in some way
 - Exerting influence on the project in some way
- Stakeholder analysis techniques are useful ways to try to identify all and all the right stakeholders

Stakeholder Analysis

- In order to work well with stakeholders the following questions should be answered:
 1. What do we require from them?
 2. What do they require from us?
 3. Why do they require it?
- Addressing these question early and proactively will avoid last minute conflicts
 - Potential conflicts are addressed early and resolved
 - Less pressure and stress
 - Helps build strong relationships

Activity 3.2: Stakeholder Analysis



Goal	Identify stakeholders for Buy Local Bonds
Team Format	Teams of five
Materials	Flip chart
Tasks	1. Discuss scenarios for developing a Buy Local Bonds application 2. Identify as many stakeholders as possible using sticky notes 3. For each stakeholder, do a stakeholder analysis using the grid 4. Can you identify supporters and opponents 5. Create a flip chart summarizing your results
Deliverable	Flip chart
Duration	30 minutes

Use Case and 3P Analysis

- Be aware that there is often a political dimension to a project
 - For those stakeholders who support the project, move directly to detailed requirements analysis
 - For those stakeholders who are opposed, 3P Analysis may be required first



3P Analysis

■ **Priority**

- What are they trying to accomplish?
- What is their current top priority?
- Do any of their priorities conflict with the project?
- What, if any, is their priority for our project?

■ **Perspective**

- Why do they have their current priorities
- Who sets their priorities?
- How often are priorities re-evaluated?
- How do they view their overall responsibility to the organization?

■ **Personality**

- What is their personality type?
- Are their other stakeholders we can leverage to assist us?
- What is the best way to approach them?

- Remember to look for the win-win!



Activity 3.3: 3P Stakeholder Analysis



Goal	Identify the stakeholders for a project
Team Format	Teams of five
Materials	Next slide, flip charts, laptops
Tasks	1. Read the narrative on the next page 2. Perform a 3P analysis using any tools available 3. Summarize your findings on a flip chart 4. Based on your research, what would you suggest?
Deliverable	3P Analysis and Proposed strategy
Duration	30 minutes

Activity 3.3:

3P Stakeholder Analysis (continued)



- Your team is promoting an initiative to implement a free bicycle sharing program in Raleigh, NC. You are now examining the various stakeholders to determine who might support, and who might oppose, this project. The project will follow the docked bike-share model used in cities such as Montreal, Paris, and New York.
- You suspect that the Mayor will oppose. There are concerns about available funding, as well as how the program has fared in other cities.
- You have managed to arrange a meeting with the Mayor to discuss his/her position
- You need to prepare your case for that meeting

Requirements in a Project Framework

Project Management

Roles and Responsibilities



Life Cycles

Managing Scope

The Requirements Life Cycle

- Requirements move through a repeating cycle, whatever development approach is chosen
 - Elicit — draw out needs through interviews, workshops, and observation
 - Analyze — model, decompose, and resolve conflicts between requirements
 - Specify — document requirements so they are unambiguous and testable
 - Validate — confirm with stakeholders that the requirements describe the right solution
 - Manage — baseline, trace, and control change across the life of the project
- The cycle repeats: every release, iteration, and change request re-enters at Elicit

Software Development Life Cycles

- The SDLC determines when requirements work happens and how much is done up front
 - Waterfall — requirements are completed and approved before design begins
 - Iterative and incremental — requirements are elaborated release by release
 - Agile (Scrum, Kanban) — a prioritized backlog is refined continuously, detail arrives just in time
- No life cycle removes the need for requirements
 - The choice changes when the work is done, not whether it is done
 - Select based on requirements volatility, regulatory constraints, and the cost of late change

Requirements in a Project Framework

Project Management

Roles and Responsibilities

Life Cycles

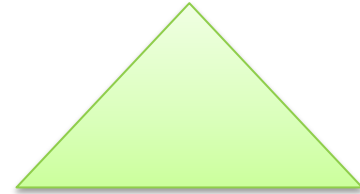


Managing Scope

Scope

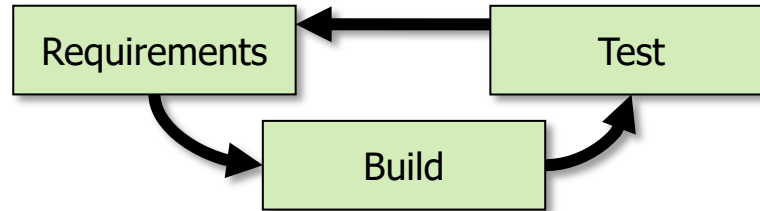
- In every project, scope is a partnership negotiation between I.T. and the business
- Scope is agreed to through a set of high-level requirements and expected deliverables when a business case is approved into a project
- Managing scope creep during a project requires:
 - Initial agreement on the scope
 - Constant comparisons of new requests to initial scope
 - Active attempts to abide by initial scope
 - Intentional decisions to expand scope when necessary

Scope = Features &
Functionality

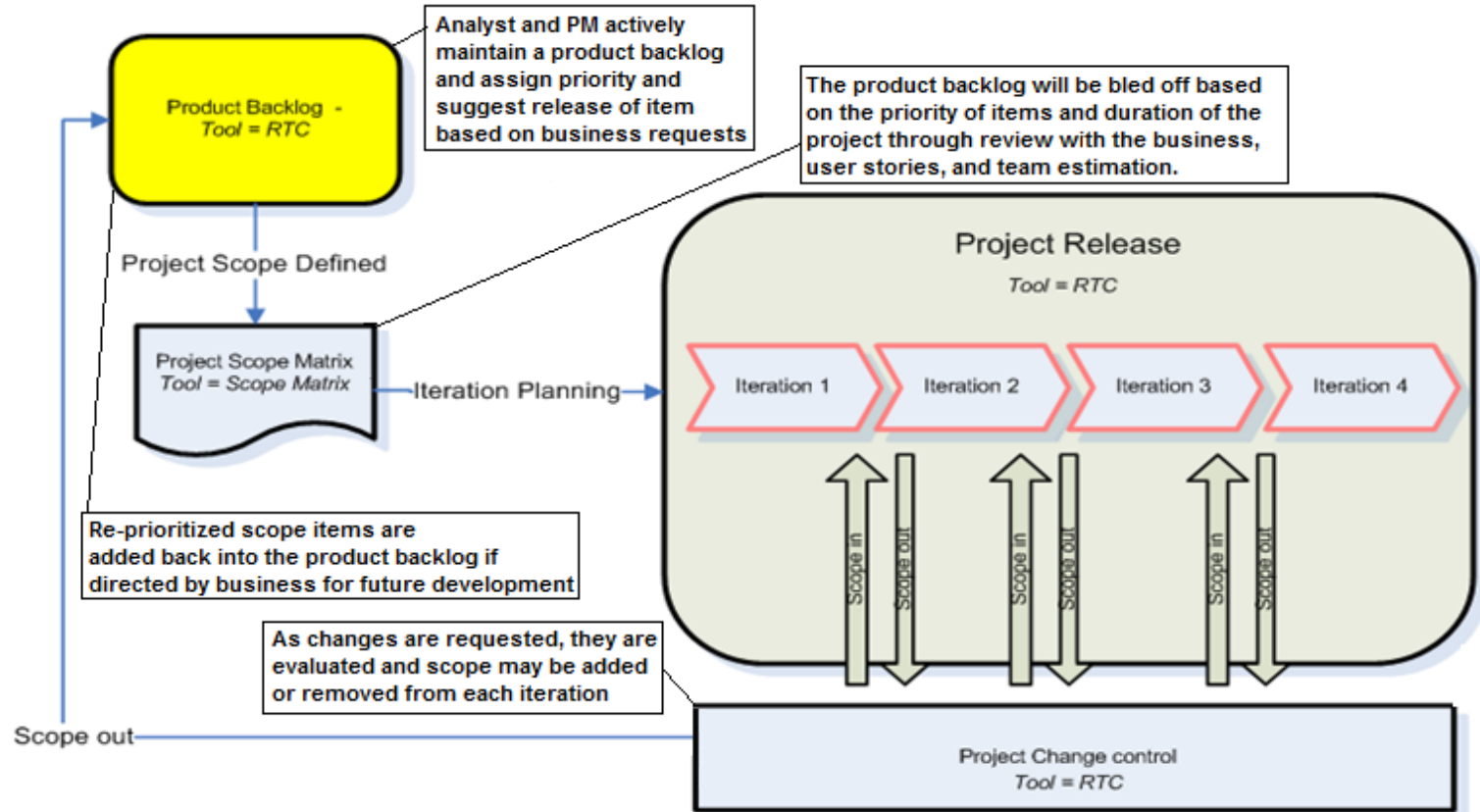


Managing Scope in Traditional SDLCs

- Requirements are allocated from the overall scope of high-level features identified for the project
- Releases are planned during the requirements cycle
 - After they are analyzed
 - Revisited after they are designed
 - Finalized before the Build decision is made
- Scope is managed by a formal Scope Change Management process



Managing Scope in an Agile SDLC



Prioritized Scope Matrix



A Workshop Approach to Prioritize Scope Items

- Attendees
 - Key stakeholders who are decision makers together with analysts
- Purpose
 - Help prioritize higher value scope items to earlier in the program
 - Keep business focused on items that were of high value to them
- Context
 - First pass at prioritization (prioritization is revisited over time)
- Prepare
 - Have high-level requirements pre-written on sticky notes
 - Divide the meeting into parts
- Conducting the workshop
 - In Part 1, key stakeholders discuss, agree, and position each high-level requirement in one of the quadrants based on the item's Business value and Technology Risk/Complexity
 - Work quickly
 - Set aside any items requiring more in-depth attention and discussion
 - In Part 2, discuss the items that required special attention
- Deliverable
 - Prioritized scope items

Activity 3.4: Prioritize Scope Elements



Goal	Prioritize a set of requirements
Team Format	Three teams
Materials	The next two slides
Tasks	1. The instructor will assign each team one of the following three roles: A. Airline Company B. Airline Technology Team C. Airline Customer (Traveller) 2. Review the categorized requirements provided 3. Discuss in your team the business value and technology risk for each requirement from the perspective of the team's assigned role 4. Allocate each requirement to an agreed to quadrant
Deliverable	Scope Matrix
Duration	20 minutes

Activity 3.4: Prioritize Scope Elements (continued)



#	Category	Requirement
1	FR	ABC Airlines shall transfer time-consuming repetitive tasks from airline staff to airline travelers.
2	FR	Users shall be able to check in at any partner airline kiosk.
3	FR	Airline shall provide automated check-in services at airports for travelers with existing airline reservations.
4	C	Airport kiosks shall not support a Braille interface.
5	NF	Check-in kiosks shall be available at airports 24x7.
6	B	The System shall assign the traveler 100 FrequentTraveler credits when check-in is performed at an airport kiosk.
7	FR	The System shall validate the FrequentTravelerID according to the following criteria: 10 characters, can be blank, alphanumeric, starts with a letter, upper-case letters, will not include the letter O.
8	FR	The System shall use the traveler's FrequentTravelerID as the primary key for all kiosk-initiated transactions.
9	NF	Traveler may check-in at a kiosk when flight departure time is greater than 60 minutes and less than 24 hours.
10	B	Traveler check-in shall be available at all major airports by the end of next fiscal year.

Activity 3.4: Prioritize Scope Elements (continued)



Chapter Summary

In this chapter, we have positioned requirements in the project framework by:

- Establishing the work involved in project management
- Establishing a requirements life cycle
- Identifying software development life cycles
- Distinguishing between key project roles and responsibilities
- Managing scope and scope change